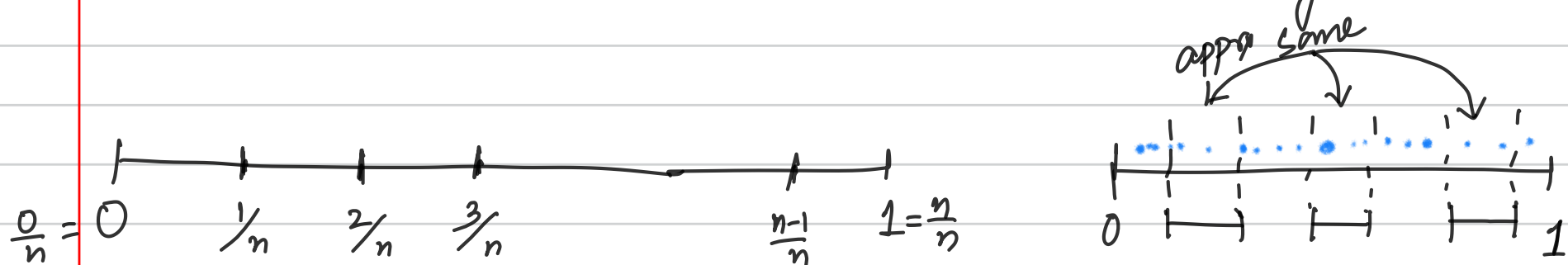


Assumption: Inputs are drawn from a uniform distribution over some range $[0, 1]$



$n \rightarrow$ subintervals

for each subintervals, we create

a list $B[1] \rightarrow [0, \frac{1}{n})$

$B[2] \rightarrow [\frac{1}{n}, \frac{2}{n})$

$B[3] \rightarrow [\frac{2}{n}, \frac{3}{n})$

$\vdots$

$B[n] \rightarrow [\frac{n-1}{n}, 1]$

p_1

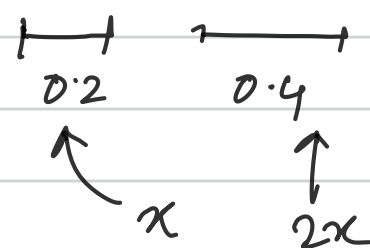
p_2

p_3

p_n

Since the inputs are from uniform dist,

$p_i \rightarrow O(1)$



Bucket Sort

$O(n)$ { for each $i = 1$ to n
Insert a_i into appropriate list

$\sum_{i=1}^n O(p_i^2)$ { for $i = 1$ to n if $a_i \in [\frac{4}{n}, \frac{5}{n})$ then Insert a_i in $B[5]$
(Any) Sort the list $B[i]$ (insertion sort)
for $i = 1$ to n print $B[i]$ } $O(n)$

$$\text{Running time} = O(n) + \sum_{i=1}^n O(p_i^2) + O(n) = O(n) + O(n) + O(n) = O(n)$$

Master's Th.

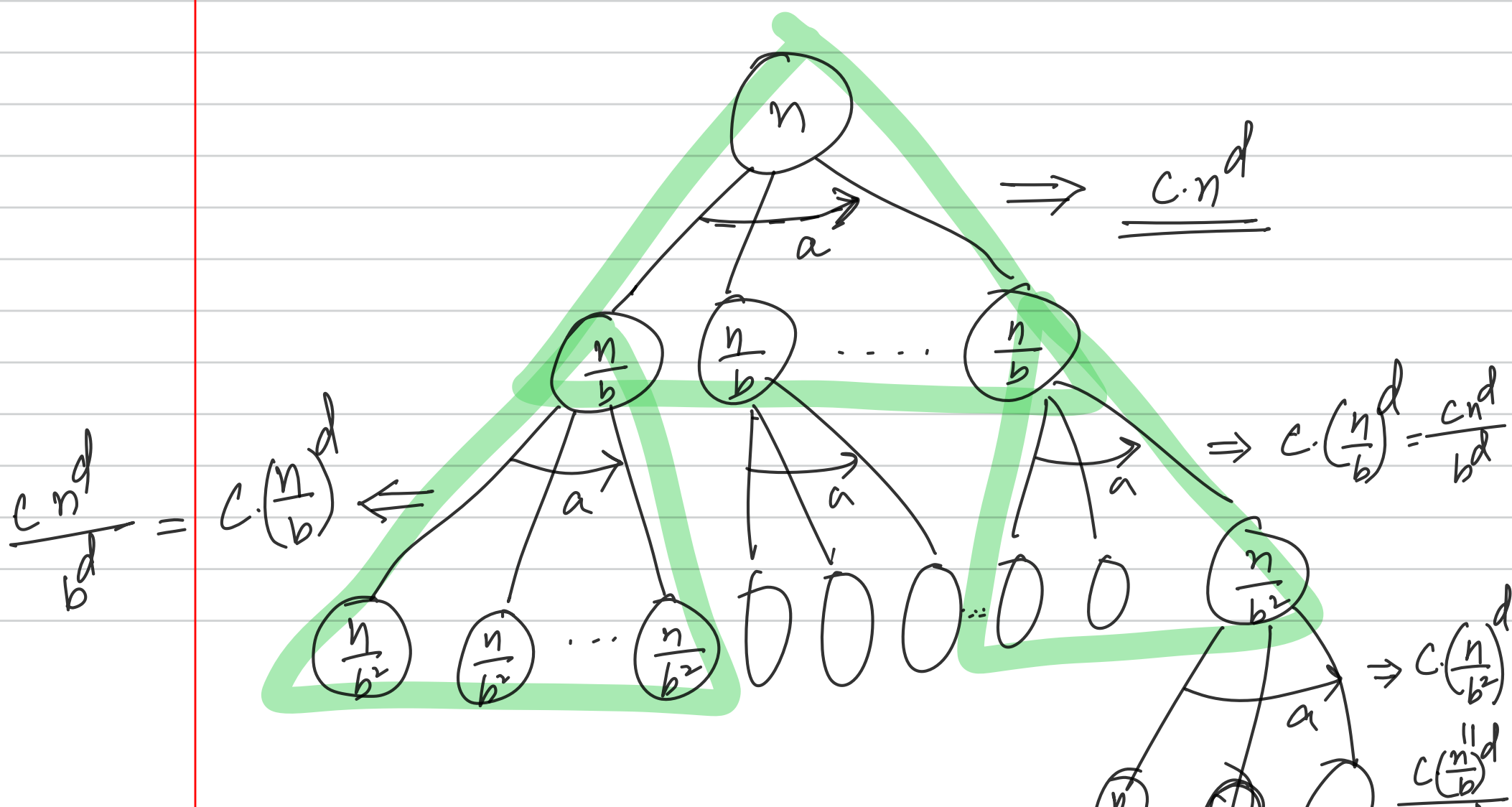
$$T(n) \leq 2T\left(\frac{n}{2}\right) + c \cdot n$$

For some div & cong algo, the running time of that algo is represented by the recurrence relation

$$T(n) \leq a \cdot T\left(\frac{n}{b}\right) + c \cdot n^d$$

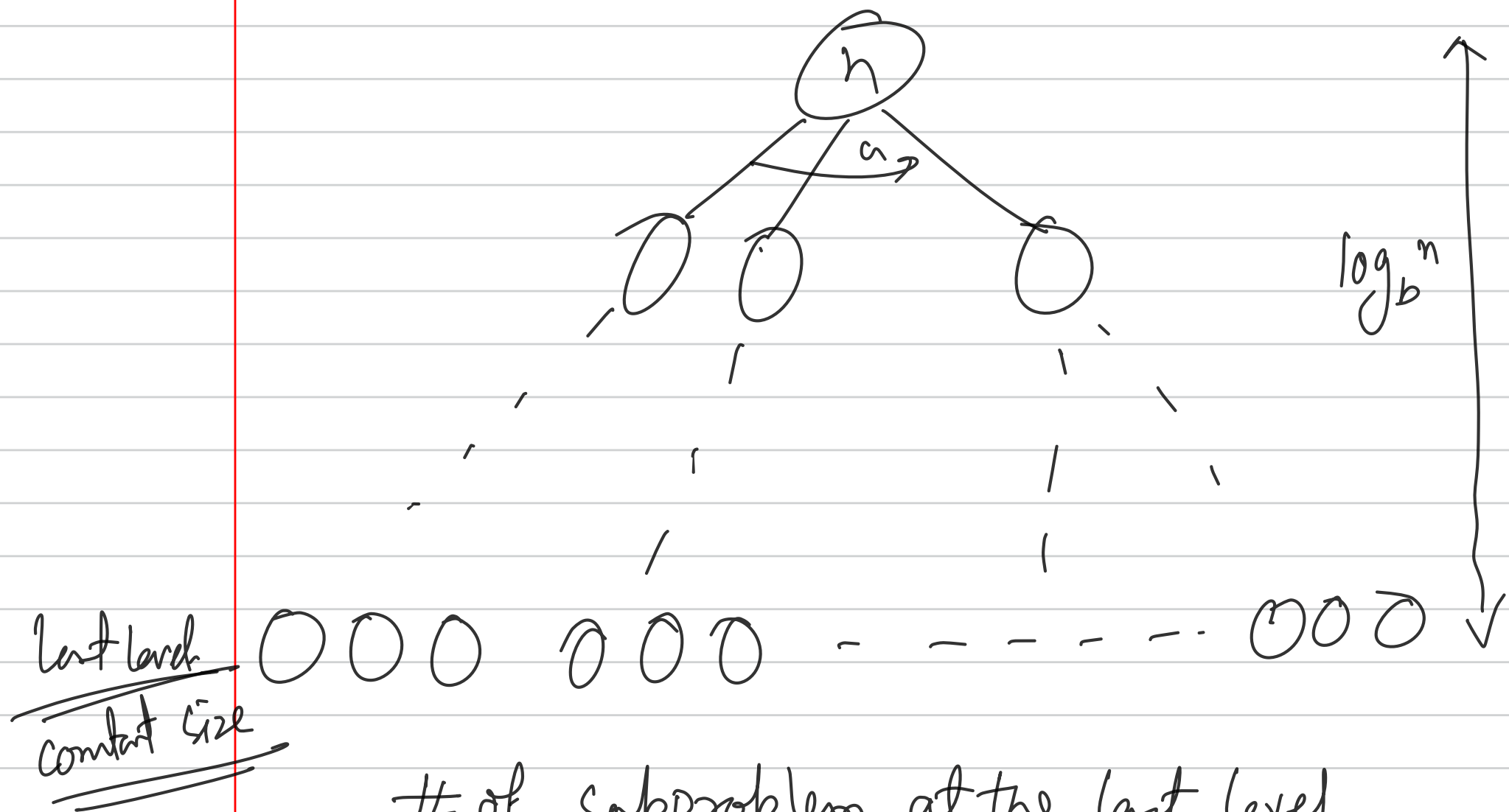
- (*) no of subprob = a
- (*) size of each of these subprob = $\frac{n}{b}$
- (*) amount of non-recursive work done on an input of size $n \leq c \cdot n^d$

$$T(n) = \begin{cases} O(n^d \cdot \log_b n) & \text{if } a = b^d \\ O(n^d) & \text{if } a < b^d \\ O(n^{\log_b a}) & \text{if } a > b^d \end{cases}$$



b^d = the factor by which non-recursive work done (Good force) is reduced.

$a = \text{no of subproblems}$ (Bad force)



of subproblem at the last level
 $= a^{\log_b n}$

and each of these subproblems has $O(1)$ size.

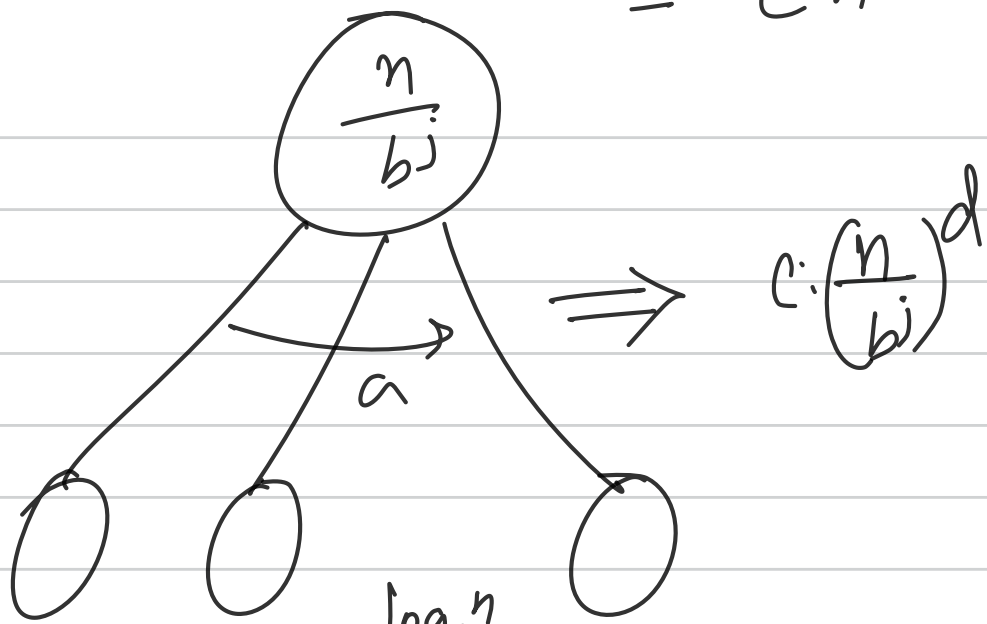
Total work done at the last level = $O(a^{\log_b n})$
 $= O(n^{\log_b a})$

Consider j -th level $j = 0, 1, 2, \dots, \log_b n$

$$\# \text{ sub probs} = a^j$$

size of each subprob = $\frac{n}{b^j}$

Total Time at j -th level = $a_j \times c \cdot \left(\frac{n}{b_j}\right)^d$



$$\text{Total time} = \sum_{j=0}^{\log_b n} \underline{c \cdot n^d} \left\lfloor \frac{a}{b^d} \right\rfloor^j$$

$$\Rightarrow T(n) \leq c n^d \sum_{j=0}^{\log_b n} \left(\frac{a}{b^d} \right)^j$$

Case I $a = b^d$ $T(n) \leq c \cdot n^d (\log_b n + 1)$

$$\Rightarrow T(n) = O(n^d \log_b n)$$

Case II $a < b^d$ $r = \frac{a}{b^d} < 1$

$$T(n) \leq c n^d \sum_{j=0}^{\log_b n} r^j = c n^d [1 + r + r^2 + \dots + r^{\log_b n}]$$

$$\leq c n^d [1 + r + r^2 + \dots]$$

$$= c n^d \frac{1}{1-r}$$

$$= O(n^d)$$

Case III $a > b^d$

$r = \frac{a}{b^d} > 1$

$$T(n) \leq c \cdot n^d \sum_{j=0}^{\log_b n} r^j = O(n^d r^{\log_b n})$$

$$= O\left(n^d \left(\frac{a}{b^d}\right)^{\log_b n}\right)$$

$$(b^d)^{\log_b n} = b^{d \cdot \log_b n} = b^{\log_b n^d} = n^d$$

$$= O\left(\cancel{n^d} \frac{a^{\log_b n}}{\cancel{n^d}}\right) \swarrow$$

$$= O(a^{\log_b n}) = O(n^{\log_b a})$$